Huang Fang

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Education

2022 - 2026 Postdoctoral Researcher, Physics, Fudan University
Advisor: Prof. Peng Tan

2016 - 2022 Ph.D., Physics, Brandeis University
Advisor: Prof. W. Benjamin Rogers,
Prof. Michael F. Hagan

2012 - 2016 B.S., Optical Information Science and Technology, University of Science and Technology of China

Research Interests

Interfacial physics and low-dimensional assemblies, including geometrically frustrated and confined soft matter and nonequilibrium complex systems. Enhanced-sampling methods for molecular simulations and AI-assisted inverse design and optimization of programmable intelligent materials.

Skills and Techniques

Polymer synthesis and characterization; optical, confocal, and electron microscopy; quantitative image analysis and particle tracking; molecular simulations with C++, Python, LAMMPS, and HOOMD-blue; high-performance computing and cluster management.

Teaching Experience

PHYS 29 Teaching Assistant, **Electronic Laboratory**, Brandeis University, 2017-2018
PHYS 18 Teaching Assistant, **Introductory Laboratory**, Brandeis University, 2016-2017

Publications

† These authors contributed equally

* To whom correspondence may be addressed.

1. **Huang Fang**†, Xiaotian Li†, Wensi Sun†, Chengxin Wang, Nuo Chen, Yining Gan, Jiping Huang, Yuqiang Ma*, Hajime Tanaka*, Peng Tan*, "Dual-symmetry-guided assembly of complex lattices", *Nature* **652**, 635–642 (2026).

2. Qiong Gao†, **Huang Fang**†, Dong Xiang, Yanshuang Chen, Hajime Tanaka* and Peng Tan*, "Impact of impurities on crystal growth", *Nat. Phys.* **21**, 938–946 (2025).

3. **Huang Fang**[†], Qiong Gao[†], Yujie Rong[†], Yanshuang Chen, Jiping Huang, Hua Tong, Zhihong Nie[*], Hajime Tanaka[*], Wei Li[*], and Peng Tan[*], "Dynamic and asymmetric colloidal molecules", *Nat. Commun.* **16**, 2819 (2025).

4. **Huang Fang**, Michael F. Hagan, W. Benjamin Rogers[*], "Two-step crystallization and solid–solid transitions in binary colloidal mixtures", *PNAS* **117**, 27927-27933 (2020).

5. Xiaoxia Li[†], **Huang Fang**[†], Krongtum Sankaewtong[†], Minhuan Li, Yanshuang Chen, Jiping Huang, Ran Ni[*], Hajime Tanaka[*], and Peng Tan[*], "Phase Reentrances and Solid Deformations in Confined Colloidal Crystals", *Phys. Rev. Lett*. **132**, 018202 (2024).

6. Quanzheng Deng[†], **Huang Fang**[†], Xueliang Zhang, Peng Tan[*], Lu Han[*], "Mesostructured Dodecagonal Quasicrystal: the Interplay Between Temperature Regulation and Kinetic Structural Transition", *Small* **21**, 2502064 (2025).

7. **Huang Fang**, Botond Tyukodi, W. Benjamin Rogers, Michael F. Hagan[*], "Polymorphic self-assembly of helical tubules is kinetically controlled", *Soft Matter* **18**, 6716-6728 (2022).

8. **Huang Fang**, Peng Tan, "Packing Problems in Soft Matter Physics: Fundamentals and Applications, Chapter 17: Packing of Charged Spheres", *Royal Society of Chemistry* 433-450 (2025).

9. Pepijn G. Moerman, **Huang Fang**, Thomas E. Videbæk, W. Benjamin Rogers[*], and Rebecca Schulman[*], "A simple method to reprogram the binding specificity of DNA-coated colloids that crystallize", *Soft Matter* **19,** 8779-8789 (2023).

10. Thomas E. Videbaek, **Huang Fang**, Daichi Hayakawa, Botond Tyukodi, Michael F. Hagan, W. Benjamin Rogers[*], "Tiling a tubule: How increasing complexity improves the yield of self-limited assembly", *Journal of Physics: Condensed Matter* **34**, 134003 (2022).

11. Simon Merminod, John R. Edison, **Huang Fang**, Michael F. Hagan, W. Benjamin Rogers[*], "Avidity and surface mobility in multivalent ligand–receptor binding", *Nanoscale* **13**, 12602-12612 (2021).

12. Yifan Gao, Shenxin Xia, **Huang Fang**, Zhebin Zhang, Dong Yang, Tongtao Li[*], Wei Li[*], Angang Dong[*], "General Synthesis of 2D Mesoporous Multimetal Spinel Supracrystals for High-Efficiency Electrochemical Applications", *Angew. Chem. Int. Ed.* **64**, e202510954 (2025).

13. Daichi Hayakawa, Thomas E. Videbæk, Douglas M. Hall, **Huang Fang**, Christian Sigl, Elija Feigl, Hendrik Dietz, Seth Fraden, Michael F. Hagan, Gregory M. Grason, W. Benjamin Rogers[*], "Geometrically programmed self-limited assembly of tubules using DNA origami colloids", *PNAS* **119**: e2207902119 (2022).

14. Wei Zhang, Wenxiong Shi, Jinxiu Wang, Kerun Zhu, Yong Tian, Haili He, Linlin Duan, **Huang Fang**, Yupu Liu, Biwei Wang, Yuzhu Ma, Tiancong Zhao, Xingmiao Zhang, Zirui Lv, Lianhai Zu, Yun Tang, Angang Dong, Biao Kong*, Dongyuan Zhao*, "A Library of Nanocrystal-Inserted Ordered Mesoporous Frameworks with Ultrahigh Density and Spatial Dispersity", *Adv. Funct. Mater.* **34**, 2401814 (2024).

To be published

1. **Huang Fang**, Peng Tan*, "Curvature-dependent mesoscale morphologies in two-dimensional crystals".

2. Yuyue Gong, **Huang Fang***, Peng Tan*, "Competing frustrations drive morphology selection and pathway bifurcation in colloidal supraspheres".

3. Hao Wang†, **Huang Fang**†, Chuncheng Li, Yining Gan, Siyu Wan, Yutong Gao, Jingxing Jiao, Xinran Xu, Siyuan Liu, Mingxuan Bian, Longfei Lv, Yifan Gao, Dong Yang, Tongtao Li, Zhaochuan Fan, Peng Tan*, Angang Dong*, "Octagonal quasicrystalline nanoparticle superlattices enabled by asymmetric attraction".

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Grant No. 2025M773335; RMB 80,000.

2022 China Postdoctoral Science Foundation, General Financial Grant (Second Class).
Grant No. 2022M720811; RMB 80,000.

Talks and Posters

1. "介尺度阻挫下的曲面二维晶体", 复旦大学物理学系 2025 年度学术年会，上海，2025 年 4 月 24 日

2. "Two-dimensional crystal on a sinusoidally curved manifold", *APS March Meeting Hong Kong Satellite Session, Hong Kong, March 18, 2025*

3. "动态"类分子"胶体团簇的非对称性", 第十三届全国软物质与生命物质物理学术会议，西安，2024 年 3 月 31 日

4. “Understanding the mechanism of self-limited assembly of tubules using Monte Carlo simulations”, *APS March Meeting, Chicago, IL, March 15, 2022*

5. “Exploring nucleation pathways and solid-solid phase transitions in two-dimensional binary colloidal crystallization”, *APS March Meeting, Online, August 18, 2021*

6. “Exploring nucleation pathways and solid-solid phase transitions in two-dimensional binary colloidal crystallization”, *Gordon Research Conference poster session, New London, NH, August 13-14, 2019*

7. “Exploring nucleation pathways and solid-solid phase transitions in two-dimensional binary colloidal crystallization”, *Gordon Research Seminar, New London, NH, August 11, 2019*

8. “Exploring nucleation pathways and solid-solid phase transitions in two-dimensional binary colloidal crystallization”, *APS March Meeting, Boston, MA, March 4, 2019*

9. “Exploring nucleation pathways and solid-solid phase transitions in two-dimensional binary colloidal crystallization”, *APS March Meeting, Los Angeles, CA, March 7, 2018*